RODRIGO PÓVOA

AI & SOFTWARE ENGINEER

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PROFESSIONAL PROFILE

M.Sc. in Informatics and Computing Engineering graduate from FEUP (18/20 final grade) specializing in applied Computer Vision, Deep Learning, and LLM-based systems. Developed an end-to-end medical segmentation pipeline in PyTorch during an 18/20 dissertation at Fraunhofer Portugal AICOS. Experienced across modern ML workflows, LLM integration and full-stack software delivery in Agile environments. I want to solve AI problems end to end, from discovery to deployment, working directly with teams and clients and learning new domains fast.

PROGRAMMING

- Python
- C++
- Java / JavaScript / TypeScript
- SQL

LLM & BACKEND

- Gemini/OpenAI APIs
- NestJS
- REST APIs

ML & VISION

- PyTorch, PyTorch Lightning
- MMSegmentation
- OpenCV
- Scikit-learn

SOFTWARE & TOOLS

- Git / GitHub /GitLab
- Docker

Languages: Portuguese (Native) and English (B2)

ACADEMIC TRAINING

Bachelor's Degree in Informatics and Computing Engineering

Faculty of Engineering, University of Porto
2021-2024

Master in Informatics and Computing Engineering

Faculty of Engineering, University of Porto
2024-2026
18/20 Dissertation Grade
18/20 Course Grade

UNIVERSITY PROJECTS

Automated Dressing Saturation Analysis

Dissertation, Masters in FEUP & Fraunhofer Portugal AICOS | 2025-2026

- Researched, designed, and implemented an end-to-end Computer Vision pipeline to quantify saturation levels in chronic wound dressings to assist clinical decision-making.
- Developed and benchmarked semantic segmentation and classification models using **PyTorch**, **PyTorch Lightning**, and **MMSegmentation**, utilizing Transfer Learning on specialized medical datasets.
- Engineered robust data augmentation strategies to mitigate severe class imbalance and limited clinical sample availability.
- Implemented modular preprocessing and evaluation pipelines, producing reproducible metrics (mIoU, Dice Score, F1) for medical validation.
- Tech Stack: Python, PyTorch, MMSegmentation, OpenCV, Scikit-learn, Git.

Zyvera / Lyra - AI-Powered App

Project Management Laboratory (LGP), FEUP | 2024 – 2025

- Built the LLM layer of an AI outfit-recommendation app in NestJS/TypeScript: a provider-factory abstraction over Gemini, OpenAI and Stability AI for text, image and multimodal generation, plus typed parsing of model outputs.
- Contributed to the Flutter/Dart frontend, integrating REST APIs, in a Scrum team.
- Tech Stack: NestJS, TypeScript, Gemini/OpenAI APIs, Flutter, Dart, REST, Git.

Convert a real 3D model into an immersive environment

Integrator Project (PI), FEUP | 2023 – 2024

- Developed a Python application to ingest, parse, and normalize 3D geometric asset data for immersive environments.
- Built algorithms to compute precise bounding boxes and percentage scale factors across heterogeneous 3D file formats.
- Structured configuration and metadata pipelines utilizing JSON schemas to ensure pipeline interoperability.